# Dr Stanislas Dehaene - Consciousness: From Computation to Cognition, Cog Neuroscience and Clinic

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and we're live excellent so welcome back everybody um we are now uh at the point where we have reached the third keynote lecture of main 2020 and it is my distinct pleasure and honor uh to introduce uh Professor Stanislas danan from college the France in Paris um so Professor danan is a professor on the of chair of and experimental cognitive psychology at the college the France in Paris and he's director of the cognitive neuroimaging unit at the or brain Imaging Center um he received his training in mathematics cognitive psychology and neuromodeling uh Professor Dan's interests concern the cerebral basis of specifically human cognitive functions such as language uh computations calculations and reasoning he has received several International prizes I won't have time to list them all here they include the MacDonald Centennial Fellowship and the the Lou de price for the French Academy of Sciences with Danielle L um I could go on and on um I can just tell you above all how excited I am that um personally and also the organizing Committee of Maine that this is finally happen happening I mean Stan probably remembers that I've been bugging him with emails every year just some months before Maine saying is it gonna happen this year so finally it's happening it's coming true we are um we have now the chance to to listen to talk keynote lecture by Stanislas danan in Maine so without any further Ado uh please join me in welcoming um Professor danan for his talk that is entitled uh Consciousness from computation to cognition cognitive neuroscience and the clinic so pleas hey hello thank you so much for this nice introduction I hope you can see my slides yes and um yeah I wish I had accepted your invitation years ago so I could be in Montreal right now but locked from my room I will try to tell you a little bit about my thoughts about Consciousness and a lot of new data uh the plan of my talk is the following first I will tell you about uh some of the contents of this first book consciousness and the brain and new data to suggest that basically I think we begin to understand the basic operation of conscious access how we perceive something in the external world we have identified signatures of Consciousness and we are progressing in this respect they are shared between animals and uh human and uh we are Guided by a theory the global neuronal workspace Theory and the data so far seem to converge rather nicely um but in the second part of my talk if I have enough time I would like to tell you a little bit about new ideas that I developed in the second book how we learn and develop the second idea that the contents of Consciousness may be different in humans they're richer in humans um we process a language of thought which leads us to Unique competencies for language for mass for music conscious representation of ourselves and others and uh we'll try to show you therefore that these are abilities that require specific models um in terms of AI this is really a neuroscience talk I think but here is where AI is going to come in first of all it's used as a tool in order to decode brain signals and to classify patients with disorders of Consciousness based on the signatures of Consciousness and second um it's used uh as a potential model I think it's achievable to model the global neuronal workspace and we've just heard from Josh Benjo about this but um I will also claim that the current AI remains Limited in its ability to capture human behavior and propose some challenges that I think are not met at least so as far as I know from simple current neural network approaches um so uh my talk will be uh based on the Progressive development of a theory of Consciousness which called Global workspace Theory started with this very nice book by G Bas which I can still heavily recommend 1988 and then Through The Years my colleagues and I have been refining the theory and if you want to read just one paper this I think this recent paper in 2017 or maybe the even more recent review uh with Peter ruar George mure je sh will give you some more details but to beef up the idea I want to start with a very simple visual illusion that I think many of you know uh here you can see um some uh check chess pieces is on the board and there's a black piece and there's a white piece and uh well where is the illusion uh I think you know where it's coming is that these uh bits of the image are actually strictly identical they're the same exact luminance I hope you can see this now if I go back and forth but we subjectively see them as completely different and um what's going on is that our brain unconsciously detects that there's a sort of Shadow sort of Gray Zone in the image subtracts it and let us see the luminance of this object as lighter because if it was in the shadow it would have to be a lighter color um what I like about this example is that first of all I think it nicely illustrates the notion that dates from helmo that the visual system operates by unconscious inference and what we see consciously is really the result of a lot of computation but also this is a subjective illusion uh it's very clear that we don't see the objective reality which would be these grade here so it's subjective and yet machine confronted with the same display would have to to come to the same conclusion if you're really trying to see the objects on the scene you have to subtract the Shadow and so subjective Illusions are not at all incompatible with even optimal computations by a machine I think that's what the brain does um so I will um use two key ideas here the first one is that there is a lot of early perceptual processing which is unconscious accumulates the evidence from the scene combines it together and that conscious perception relates to a later wave of longer lasting neuronal activity that integrates the incoming signals for instance here it has to integrate the knowledge of the Shadow and illuminance in order uh to create more Global and globally available information so this is of course the idea of the global neuronal workspace I've shown this picture many times uh what's the idea first of all that there are many chains of processes in the brain cortical areas are typically autonomous processes neurons operate in parallel and all of these chains can operate unconsciously and there can be many such chains at a given time but the second claim is that what we subjectively experien as Consciousness is the global availability of information there are some chains the green one here that end up in this high level system for sharing information and at this in turn they can contact other processing chams link them together and the idea is consciousness relates to the activity of This Global neuronal workspace that evolve to select a piece of information from this chain for instance here so to attend to it to broadcast it to other systems leading to a Global Knowledge inside the system and the ability for instance to report verbally the stimulus to act upon it to memorize it and so on and so forth in terms of physiology the hypothesis has been that conscious access correspond to the ignition of neurons with longdistance connection you need these long distance connections to broadcast information that would be distributed in prefrontal cortex and other associative C cortex this is not a localist theory of Consciousness but the idea that prefontal cortex is just one of the nodes here and that top down signals are also being sent back to the processors and I won't go through that but there's more and more evidence for this notion of a sort of central system a bottleneck of multiple areas that speak to each other very quickly and for instance in the work of Henry Kennedy here from anatomy of the mountain so what is common to all conscious processing States well not necessarily the same no not a single state or a single marker or NCC but more a processing style a type of neural trajectory that involves this broadcasting operation and so what are the predictions of these models um I speak mostly about the predictions about the trajectories of conscious and unconscious processes and that's why I will be using in my Talk data mostly from Meg and also neurophysiological recordings rather than fmri because we want the Dynamics so the Dynamics first of all is that there should be propagations of activity coming from unconscious ston they may not reach Consciousness but they can still proceed along deep processing lines second prediction there should be this sudden Global ignition all of these areas come together this create a nonlinearity in the system Global amplification when a novel content gains access to Consciousness and uh this should be seen at the level of prefrontal Cortex neurons among others they should contain this is a very strong prediction there should be a complete neural code for any conscious content at a given moment in the prefontal cortex and related areas there are two other predictions this is a bottleneck so only a single conscious object can make it at a given time so if it's the green one it's not the orange one for instance and there's also the idea of spontanous ignition you don't have to have an external stimula it doesn't have to come from the outside from the periphery it can come from inside from memory from spontaneous patterns of thought um but I won't have time to cover all of these points so I will focus on this of sudden Global ignition coming after a series of processing stages that are unconscious we've been doing a lot of experiments with this and uh the typical experiment goes like this we will flash an object for instance a digit here it's flash for a very brief duration and then there is a mask and the mask can come after a variable delay which we vary here and if the delay is very short you don't see the stimulus and you don't even distinguish it from Target absent trials at threshold sometimes you can see it some sometimes you don't and above threshold typically after something like 60 70 milliseconds you can see the stimulus and we've been chasing in the brain the activation caused by such stimuli this is an old experiment already from 2007 only using EG and we could see that there are indeed several waves of processing what we could see here is that if you record from visual cortex you see a first wave which is essentially linear in activation as a function of the evidence provided by the stimulus before being cuted by the mouse so you can see evidence accumulation here and then there's this second wave which by the way is the only one which is seeing very strongly here in the prefrontal cortex following this sort of slow buildup here there's this nonlinearity in the system and here the curves are separated much more nonlinearly for instance if you can see if you can focus on the green curve you can see that a 33 millisecond stimulus will create visual activation but then it dies off and uh there is no second wave here what we could see is that the fraction of C digits the subjective reports of the subjects correlates with the size of this late wave and we propose that this is a possible signature of Consciousness I just like to point that this division between a linear phase and a nonlinear ignition phase uh is present in many many experiments and we have even been able to use it to find that there is such an organization in the baby the baby cannot report but we can still see that if you flash a phas for a short duration this experiment was done by S Creer um uh there is a linear variation in the evoked potentials as a function of phase duration here and then a nonlinear stage where only the long duration phases create this sustained negativity whereas the short ones at essentially Return To Zero so we believe that there is ignition already in the young baby uh at certainly at 12 months we see so it very clearly and I think this is the data from 12 months but perhaps much earlier probably maybe even very close to birth the earliest evidence we had in this experiment was at five months so it's a basic architecture of the system we can see it many times we have a new tool now to look at these things which is multivariate decoding and this has been a very strong contribution of je King who did his PhD in the lab um what we can do is we can take a slice of time coming from these uh EG Nets or from the mg that you can see here and using all of these sensors and this slice of time we can try to build the decoder for what the subject has been presented with for instance decoding which image he has seen is it a face a house an object or a body part and this is actual data showing that of course before the stimulus you can't decode after the stimulus there is a period where you can't and then you can decode the stimulus but what's great about this is you can also use the pattern of generalization across time to illuminate the organization of the system think about it if there's a chain of processors that they each have a specific code then you could train on a given time and you could decode at the same time even with new data but you could not decode at a different time because the activation is gone onto a different code but if there's a stable sustained ignited code then you could train at a given time and still generalize to a different generalization time using different data from a different time point because the data is stable and it's not just Theory because this is what we see in the actual data so this was the work of Sebastian Marti you can see here when we flash a picture we do see a sort of diagonal pattern of decoding which means we change a given time we can mostly decode at about the same time and then the information moves to another coding scheme but later in time you can see this slate much more stable pattern and I will outline it for you here so you can see there this much more Square pattern which means that the neural activity has become stable in time and we can decode it and it seems in our hands that this diagonal phase corresponds much more to non-conscious processing and this more stable phase to conscious processing so one way to see this is to use this decoding in this masking experiment so here is our masking stimuli what we decode here is just the presence of the stimuli can we distinguish between a stimulus which is present like the D nine here and a stimulus where there was no digit at all just the mask we can make this distinction we can make it even when the stimulus is extremely short like 17 or 33 milliseconds here and this is a subliminal stimulus below the threshold so you can see that it creates a whole pipeline of activation but notice what happens when we cross the threshold here there's is much more intense activation building up and then suddenly there's this intense Square pattern stable pattern of activity which if we simply take the same exact stimulus and separate unseen thec trials nicely separates uh these two categories you can see that there's much more stable activity on the scene case and this has been proposed as a notion of metastability it's not that the activity is no longer changing it's still changing but it's changing much more slowly the system has reached a much more stable state of activity which we claim is associated with broadcasting um je Remy has been replicating this pattern many times initially our experiments were criticized because we were using symbolic digits so we did them again with grids you can see here and in this experiment also with a long delay before the subject sees a probe and has to decide whether the probe is tilted left or right compared to the target you have to make this Choice even if you don't see it it's very interesting that you can be above chance so even after a long delay like this of close to one second um you can still use some of the subliminal information provided by the Target but of course you're much better when you cross the threshold the information becomes much more stable in your mind when we decode this situation we see once again a sort of diagonal pattern for the Unseen trials here so we see a trajectory in neural space and surprisingly we can continue to decode the Unseen stimulus for a long time but still you can see that the non-conscious trial is evoking decaying activity in blue the SE in stimulus is evoking much more stable activity sustains whole time and uh even though it's not a perfect square it continues to evolve through time it's much more stable much more broader in this Matrix compared to the more diagonal pattern evolved by the Unseen stim creating this difference here so this is what we call ignition it's not just a stable pattern and it's a stable trajectory and this notion that conscious and unconscious processing can be seen as divergent Dynamic trajector has been a point that has been made by others for instance Mo salti also B what they see here is the same as we see the seen trials even for a fixed constant stimulus the seen trials diverge very quickly from The Unseen trials if you measure the speed of Divergence I saw this was a very clever trck or just measure the speed with which the magnetic fields the mg change you can see that suddenly they start to change very quickly for the seen trials compared to the Unseen trials that's what we call nonlinear ignition um now can we go beyond this simple observation of ignition to what it does what's the computation our idea with Jean REM and we proposed that in his paper in 2014 very formally is that uh the computation is one of the posterior think of my original example you have to compute the posterior distribution of light uh the probability that this is white or dark as oppos based on all of the information on the seene so it's a perceptual decision following a long unconscious inference uh we wrote this sentence which I still like what we call a subjective report May simply be the brain's best attempt at solving a difficult perceptual decision problem with myriads of potential classes each with different costs and prior probabilities that depend on the subject's prior experience so the subjective report is not random at all subjective is a computation but a computation of a posterior in order to test this idea je and Gabriel did a very recent experiment it's not published yet but uh the idea is that we can manipulate the evidence by masking but in order to separate it better from the posterior we can also manipulate the prior so we have a masking Paradigm here we flash a digit there's a mask we vary the S SOA the delay between the digit and the mask but we also vary the prior here by having two blocks you can see them here there's one block where most of the trials are invisible because we present all of these short duration trials most of them are invisible and your prior is that you're going to get a trial where you don't see the stimulus and then in another block we present some of the same durations but also longer durations so that the prior is that you're going to see most of the TRS and uh if you look at the behavioral data this is sufficient shift as predicted by the model the subjective reports the the subjective reports of seeing integrate the prior and shift uh the second block towards more visibility even though it's the same stimulus and even though the objective discrimination as predicted by the fact that this is based on unconscious computation the objective discrimination Remains the Same if we look at what happens at the mg level we once again replicate this idea of a trajectory for unconscious stimuli which gets broader uh for when the stimulus is seen and if we contrast unseen versus seen stimuli simply based on this middle stimuli where the stimulus is the same and one of these durations here so for a fixed s SOA uh we have this distinction between seen and unseen stimul which creates this region of significant difference here amplification of the activation followed by this very late metastable phase um but now we can dissociate it from SOA with can do multiple regression this is what we what jean Remy did here to predict the outcome of this detector for present versus absent trials and we can see that the early stages are dominated by the SOA by the objective evidence the duration of the stimulus before the mask and uh this is the visibility the subjective component whether the subject declared that he could see it and we can clearly see that these affects different stages of processing what about the prior we can also identify a modulation by the block uh with which the subject is confronted is it a visible block where most of the trials are visible or is it an invisible block and this prior affects the entire trajectory of processing but primarily affects the beginning even prior to the presentation of the stimulus as should be expected perhaps from the term prior so this is the key figure if we put together all of these influences we can see the successive effects of the prior of the evidence and of the posterior the visibility ratings um on uh the decoding of the presence versus absence of the stimulus and I hope you can see these chains of processing which go uh through time here each of these is a slice of the decoding so this is a decoder trained at 120 milliseconds to decide whether there was there was or there was not a stimulus and you can see that the decoder is first influenced by the prior then by the evidence and finally converges to the posterior and we see that multiple times I think this is consistent with the idea of a chains of chains of processes that all have the same sort of computation the same style of computation at different levels and eventually when you reach the last level here by 300 millisecond you're completely dominated by the visibility the subjective final posterior of the subject now all of these Meg data is very nice but as you can see I'm showing it to you in decoding space we can also do s modeling and we can see that prefrontal cortex contributes to this and this is a distributed response but the best of course is to go to animal research uh in order to be able to pinpoint the neural bases of this ignition and so uh we started doing this first of all in collaboration with Peter ruin in this recent science paper we had a monkey model of a flash stimulus Crossing or not crossing the threshold so this the param was even simpler than before we just flash a little bit of light with a variable contrast and sometimes there's no stimulus at all so zero contrast and uh if the monkey sees the stimulus he has to make a sad to the item and if he doesn't see the stimulus he has to make a sad to a fixed dot or to report not seen and once again we have Behavior which is characterized by this by this nonlinear thresold function there's a thresold contrast for a given animal uh below which you don't perceive the stimulus even though it's present and above which becomes very easy to perceive and there are some ambiguous stimuli here so we could ask in monkeys how does the firing of cells in V1 V4 and prefrontal cortex varies as a function of seeing or not seeing including for the first time here the presence of a large number of Mis stimuli that is to say stimuli that are present there is a contrast on screen but the animal reports not seeing them and what we saw is very clear I think you can see it by yourself the visual areas V1 and V4 can cannot be the basis of conscious perception because they can be very strongly activated both by sinen stimuli but also by Mis stimuli so it doesn't really matter much to these areas whether there has there's going to be subjective perception of seeing or not uh on the other hand the prefrontal cortex as you can see here um ignites whenever the monkey reports perceiving a stimulus it does so on seen trials you can see the green curve here um and it also does so on false alarm trials where there was no stimulus but the monkey thought that there was a flash of light and made this um most important I think is the Divergence between the red curve and the green curve here they're the same stimuli but um in one case you missed the stimulus the monkey uh reports not seeing the stimulus and you can see that there is what we call failed ignition there is a beginning of a wave reaching prefontal cortex but then it converges to the low state which corresponds to the one when the monkey reports not seeing so we really have a sort of dynamic system here diverging between two possible States the high State for the monkey reports yes I saw the stemulus and a low State this is just about reporting ver just not reporting so is really sort of neural correlate of signal detection Theory here um but uh according to the theory we should be able to go further than that and really we should see that there's a detailed neuronal code for the consciously perceived contents and that this has nothing to do with report it should be seen even in no report paradigms where the monkey doesn't have to do anything so this has been a beautiful work by FIS panagos first in Nikos logothetis lab which really pioneered all of this work on Consciousness in monkeys and then now in neurospin in our lab um so what he's doing is using a no report Paradigm binocular rivalry situation where the nice trick is that if the rivaling stimuli are grids and in one eye they're moving up in the other eye they're moving down the two stimuli rival so even if you're a subject you're a human subject you see one stimulus and then you see the other they the stimulus is constant but your subjective perception alternates but you don't have to ask for a report because you see it in the eye movements the eyes track the visible the visible stimulus and if you perceive movement towards the the top of the screen for instance you see slow movement followed by nagus to return to the initial position and then you know that the subject is seeing the top stemulus so using this trick FIS has been able to show that there are neural populations in the prefrontal cortex that encode which stimulus the animal is seeing even when there is no need to report at all and in fact even when there are no eye movements at all just these uh states of the prefrontal cortex switch and they switch in correspondence with the perception of the animal whether it's physical switches or whether it's binocular rivalry and the switches are purely in the head uh purely subjective there is no no switch in the actual stimulus now this is just for up Motion versus downward motion so you could argue this is a very poor um contents of Consciousness the contents of Consciousness should be much richer than that so can we go to much stimuli well that's what joim B and FIS are doing at neurospin we have these uter arrays that allow to record from you know uh close to 100 neurons in the ventrolateral cortex of the monkeys and uh we can see what happens when you just submit them to a passive viewing Paradigm where they see an object and can we use uh the firings of the cells in prefrontal cortex to decode what the monkey is seeing well yes we can and not only that but we can decode every single picture that he is seeing out of a large set of pictures such as 10 or 15 so there is a neural Vector that suffices to provide a lot of information compared to chance about the identity of the picture and we can decode each of the successive pictures even in a relatively fast tram where the subjective impression is that there is a very fleeting view of the image if you're just submitted to this as a human subject you say well I can see almost every picture they are going very very fast they keep changing each other what we see in the prefrontal cortex if we look at the decoding matrices is this sort of very short diagonal followed uh by cutting by the next stimulus so for the last stimulus you can see there is longer decoding and the Matrix is actually longer and leads to this more stable State here but we also have the same state this is the same decoder in the horizontal slice here so the same brain state allows to decode the identity of the picture only for a shorter duration when the picture has been presented inside this visual stream of 10 pictures per second or rather five pictures in half a second so what that means is that this fleeting impression of seeing a picture has a correlate in the prefrontal cortex which can be decorated um of course what would be nice would be to be able to ask the monkey to hang on to one of these pictures here is just pass of these pictures flash by and you can sort of briefly see them what would happen if we had to select a picture hang on to it and report it later it's very hard to train in a monkey maybe we will be able to do that later but we did that in a human again with Sebastian Marti and uh so this is the Paradigm you can see all of these pictures being flashed one after the other in this case there's nine pictures per second it's a long screen of 12 images um and uh we present a green frame around one of the picture and the subject has to decide what was the identity of this picture keep it in mind keep it in the global workspace because you have to decide among all of the possible pictures which one you saw when we do this we can build a decoder based on Independent data where the pictures are presented alone this is what I showed you earlier we can build a decoder for which picture is being presented is it a face is it a house is it a body part or is it an object but then we can apply this decoder to the stream of pictures and the result is similar to the monkey result I showed you earlier we see these streams of decoding we see this decoders that for a certain duration of time um are able to detect what was the identity of the image one then the image two then image three and so on so forth coded by color here so I insist that one orizontal line through this graph is one decoder but the decoder spits out a different answer that corresponds to the image number one then the image number two then the image number three and later in time you can get another decoder that decodes the the subsequent image so an interesting aspect of this data is if you take a vertical slice through the data here you have the same exact data point but it can be decoded in different ways if you capture the early activity you're decoding the image let's say number seven but if with the same data use a different decoder you might be capturing the image six or the image five you're capturing the previous images so we we must have this idea of a brain which is constantly crisscrossed by these bottom of perceptions coming into the system and reaching into prefrontal cortex now what happens when one of them is a target so now we can make the same exact decoders for Target versus non-targets and what's the difference well we see essentially two differences there is an amplification of the late activity just like you should have expected from the previous experiment I showed you uh the target is held for a longer duration so now we're reaching 500 600 milliseconds and you can still decode the the the target you can no longer decode the non-target here and at the same time there is an amplification of an earlier decoder around 170 millisecond which used to tell you what was the image but now becomes Amplified the second time at the same time as the top so this is probably top down amplification what's very interesting is that these two decoders decode slightly different things the late one decodes just the identity of the picture which is your Target and when a subject makes an error it decodes the error but it's really amplifying one particular item it's discrete all or non selection correlated with subjective reports so we think that this corresponds to the access to the global workspace the conscious report of one particular item and the others you forget but at this earlier time you see that there is partial and gradual selection there are several candidates that's normal because there are as I told you at a given slice of time here there are many possible candidates in the early visual areas you have one picture but in the later areas you have the previous picture in this pipeline so this earlier um activation corresponds to several potential candidates but at the later stage here there's only one is being selected and that can enter This Global workspace and be report so I think this characterize a little bit the architecture of the system pipelines of unconscious processing making infuences and then a single selection system um I want to move to an auditory Paradigm to tell you that the contents of Consciousness can be even richer than what I described to you already um because of course it doesn't have to be visual it could be auditory but also in the auditory domain it's not just about the identity of the stimulus but it's also about the organization in time and we've used this local Global auditory novelty parano to investigate the representation of stimuli through time so um I hope you will be able to listen to the [Laughter] stimulus okay so this is our basic Paradigm we have you listen to a sequence of sounds and in this case the last one is different and this will generate mismatch responses in your brain the soal mismatch negativity and the p300 waves you are surpris then you attend to the stimulus because there is a surprise here but then we adapt we adapt you to hearing this Global sequence of five sounds Bo and we do that several times and at the end you still get the mismatch negativity but you lose the P3 this is what we call local novelty alone globally you expect the entire sequence of course your early unconscious processor still detect that there's a change of tone but that's expected consciously and now what happens is that if we present you a monotonic sequence of tones that's when you react and you say oh I'm surprised there is a global surprise the sequence is not the one I expected so in this way we can dissociate the local transition probability from the global surprise we can have two levels of Errors if you want and in this case the error is that there is no error okay so it's a hierarchical error process processing paradig we can view it in this way this is the full design where there's one block where this is the base stimuli and this is the surprise and there's another block where the base stimul is monotonic and this is the surprise so we dissociate local surprise from Global surprise now when we do this we find two completely different types of computations on top here you see the local effect local in time as well as in space it's a very fast response it's this mismatch response it primarily sits in the auditory cortex although there's also a little bit of prefrontal Cortex ability activation it's very sharp response you can see it in intracranial recording you see the first four sounds they habituate they become smaller and then suddenly you recover this very high activation even bigger than the beginning for the Deviant SS and it's a completely diagonal pattern of decoding which means it's a sequence of sharp processing steps for this stimulus and we find it can be completely unconscious you can keep that even if you don't attend if you asleep if you're in you may still have it the global effects completely different is distributed in the space of the cortex it's much slower typically after 300 millisecond 200 300 millisecond and it creates this very broad decoding ability it's a sustain state if you decode it at a given time you can see it at lat times and we've seen that this one disappears on the unconscious situation we've explored a lot this Paradigm to show this dissociation between the local unconscious versus global conscious response and we've we've done it in the monkey as well this has been done by B Jara and Lin URI in the lab we've been able to do fmri with the monkey to see that there is a global workspace in the monkey as well there's this distributed system of areas that come up only for the Global Response and in recent re research we've seen that the Global Response tend to disappear suddenly during anesthesia when we fall asleep regardless of the type of anesthetic and will reappear and this is perhaps the most interesting aspect from a clinical point of view it will reappear if we stimulate uh using deep electrodes the central medial nucleus of the thalamus this is still unpublished work by Jordi tasi I hope soon to be published so what we have here is a system that is essentially a reflection a signature of Consciousness if you have this novelty response at this high level of processing which integrate all of the five sounds together to say you know this is my expected Melody and it's been violated then uh this is a conscious level of processing and you only get this response if the animal is conscious now now that we have this param we can go back to our electrodes and ask what is being coded uh in the prefrontal cortex of the monkey should can we see these Global responses can we see a code for the for what the monkey is seeing so you know that with these U arrays we can see several single neurons and uh this setup has been provided by F Panago te and this is the beautiful work of Mari B here so what what do we see here first of all we see that we can decode from prefrontal cortex neurons the identity of every object in a sequence we've moved to a visual presentation because we record more neurons that respond to Vision in the monkey but this is a small difference so bear with me so we can see the identity of the stimuli here blue blue blue blue orange orange orange orange Etc we can decode the identity of the stimulus I've already told you this so if the last item is different well we can decode that the identity of the object has changed but in fact we can decode every single bit of knowledge that the monkey has of the sequence and that's exactly what I told you at the beginning there should be a code in the prefrontal cortex for every single bit of information that the monkey has every single bit of conscious information so I told you we can decode which picture just appeared is it picture a or picture B um this is running blocks so you can get decoding even before the stimulus appears but of course it gets Amplified and we don't have to run this experiments in blcks we can decode identity but we can also decode the ordinal position there's something clicking in the monkey's head and in our head when we go through the stimulus that says well this is the first sound this is the second sound this is the third sound and when it comes to the fourth sound maybe I should expect something different uh because it's the end of the sequence and I know it's going to end differently so we can see this we can see neural codes that have first been described by Andreas nether that care about the ordinal position of the St um because we don't vary the SOA for the moment and it is again unpublished data we cannot really tell whether this is number of time but we think it's a mixture of both and also some notion of first and last in the sequence you can see item one item two it three four there's a mixture of identity information and ordinal position information that tells you where you are in the sequence and then there's much more than that there is is the last item different if the last item is different Fates disuse mismatch response which can be seen in prefrontal cortex and of course the monkey knows consciously uh that the last item is different and uh interestingly we also see this numerical response here you can see something that says oh it's like the first item it's like a new group of one this light item is different so it's a little bit like one okay um we can also decode the global deviant so we have other neurons that allow us to build this decoder here which is expectedly slower in time and tells us did I just hear an unexpected sequence I expected that it should end with a different sound and it didn't so I get this Global Response here so there's something in the prefontal cortex that says I'm surprised I should work more and what's very interesting is we even find neurons that before the onset of the sequence and code what should I expect the monkey knows as we know in this Paradigm that he is in a block where it should end with a different sounds so we can decode that from the beginning from specific cortico populations so all of these neurons are intermixed together we can think of prefrontal Cortex as having multiple vectors that each code for these dimensions of experience but the entirety of experience is reflected in prefontal cortex and I insist that this is in a non-reporting mon there is no report here just passive listening to these sequences so I promise to go very quickly to the clinic just to say that this local versus Global Response is just one of the tools that we have Curr in the clinic uh to decide whether to help decide whether a person is or is no longer conscious there is very difficult uh clinical problem of deciding whether someone with in a vegetative state or minimal Consciousness So-Cal unw sorry vegetative state is now unresponsive wakefulness syndrome uws here or minimal Consciousness MC how do we decide it seems that some of these patients in apparent vegetative state are actually still conscious where we can have them listen to this Paradigm and as shown by jacobo this is one indication that they might still be conscious if they still get this Global Response the local response we know we can get from comos patients it's not the only tool and from Global workspace Theory we have several additional tools we can simply look at the amount of sharing of information especially in the Theta band here in the long distances of the cortex the more there is sharing of information the more likely the patient is to be conscious and we can also look at fmri and this is something I won't have time to describe at all but just the resting state activity in fmri has become a very interesting cue to the state of the subject is if there is variation in the resting state activity as you can see these matrices of functional connectivity varying through time then there is the high likelihood that the patient is conscious this is a signature that we found in the monkey first and translate to the human so by combining these signatures we can begin to have automatic classifiers of patients um using multiple EG markers uh we can classify much better than chance whether a patient is or is not still conscious I will not have time to describe all of the specific measures here but from a clinical point of view it does not matter of course it matters from the theory what are these signatures but from the clinic the important thing is that we can begin to help classify these patients better and very interestingly inside the vegetative state it seems that uh we don't always agree with the clinical classification which is provided by the doctor the classifier the machine learning may not agree and in fact the machine perhaps predicts better at least sometimes because a greater number of patients that are classified as being conscious tend to recover as opposed to not improving so there is also a pronostic value from some of these markers this has become quite an industry of decoding of these patients it's all done in the laboratory of lonel Nash and in this paper putting together a lot of data from different groups we show that using random Forest algorithm we can get robust EG decoding uh and what's important is that it is cross sight now so from different clinical sites maybe using different EG machines we can begin to have a common classifier in the last part of my talk and it will have to be necessarily a bit short um I would like to uh move to this theme that I mentioned at the beginning do we all have the same contents of Consciousness I hope I've shown you that from makak to human we have similar mechanisms of access to Consciousness when we perceive a line or when we perceive motion or perhaps a face the mechanisms are very very similar and we have this nonlinear access to Consciousness but my hypothesis is that there is something different in a human when it comes to higher level States we have richer States we are able to represent information in a deeper manner in this review in the journal neuron we discussed the representation of sequences with this idea coming from this local Global Paradigm that sequence knowledge is already present in the monkey and of course the monkey already knows about the transitions between the items and knows that there's one which can be expected I've shown you this other experiments show that the monkey can group the items together into chunks the monkeys also know about the order of items this is the first this is the second this is the third I've shown you that and the monkeys may even be able to learn an algebraic pattern so the first two are identical for instance a a b a a and then there is perhaps a violation of this pattern all of this is available to monkeys and probably many other levels as well but there might be a level which is unique to humans this is the level of nested symbolic structures in order to describe the structures of language such as this very simple phrase here those gifted car factory workers you need three structures you need parenthesis you need to be able to say it's a Car Factory and then there are workers in this Factory it's not just car factory workers as a sequence of words it's a tree and the same applies to mathematical patterns or mathematical equations to musical patterns or to theory of mind we need recursive structures that are nested inside each other so our hypothesis is that this may be unique to the human brain and that the human brain uses these three structures to further compress the information to discover some of the laws symbolic laws perhaps that uh Joshua bjo was talking about uh it uses nested three structures to further compress the information into extremely compact Expressions I don't think I have time to show you all of the Power of these ideas so I'll refer you to this book how we learn but I think this is a key idea for how humans grow Concepts and I'm very influenced by the work of Josh Tenon bom and in this paper by Kemp and Tenon bom they show how to build an algorithm that can build new Concepts out of the grammar generative grammar or smaller ones and build Concepts such as the tree of P through this idea of a sort of language of thought a grammar of thought so we're starting to investigate whether this actually occurs in humans and not in the monkey using extremely simple paradigms I will show you one you'll be surprised by the simplicity so I hope you can see the movie on the screen you can see that there are eight locations here and this is a test for young children we tell them there's a fish which is hiding in the spawns and U try to guess where he going to go okay so this is one trial learning I hope you guessed where it's going to go next everybody says he's going to go here and then here and then here okay you've guessed the pattern even before it occurs try to do that with neural networks you know it's not so easy this is really very fast learning and what we've shown is that in order to understand this sort of learning you need uh to assume some kind of language of geometry you need to assume Primitives for instance you need to assume that the subject understand that all of these diagonals they are all symmetries on the fixed axis they're all similar in a certain sense and so you can anticipate that this is the next one and then there will be this other segment here um and you need a sort of formula that expresses this repetitions this formula is a bit complex but really what it says is that there is a repetition of four segments there's a first segment second segment a third segment and a fourth segment here and you can build other shapes such as two rectangles as long as there is regularity you can find an expression which is Compact and captures the regularities that human captures well we've done a lot of work with this I won't bore you with all of the results but just to say that our memory is driven not by the length of the sequence in fact all of the sequences we used were always eight items but it's driven by the minimal description length which we call Cog complexity Al the lengths of the shortest expression that can compress the sequence and if you don't have a language you don't have the correct expression for these minimal description legs it's not just slots this idea of working memory as slots but I think in humans it's not just slots it's really compressing the information um well we have a lot of ongoing work at the moment to try to see if this is true in the monkey as you can see we did experiments with adults we did experiment with preschoolers so it's doesn't have much to do with being trained in mathematics we even train tested adults from the Amazon that have very little access to education and again found that you make more error for sentences for phrases that sequences that are more complex that is to say they have they are less compressible they require a longer expression in order to be described well this relationship we don't find in the monkey at all in the current work which is done by leing Wong at The Institute of Neuroscience in Shanghai after staying in my lab for many years uh we see that monkeys do not seem to care at all about the temporal or geometrical regularities they just tore the locations in working memory they don't seem to care for the structure of the transitions in order to understand this language you need to care very much about the nature of the transitions much more than the particular locations you need to be to be able to understand that it's four segments regardless of how they are oriented for instance and monkeys don't seem to do that in very recent work which has just been accepted in plus computational biology we extend this approach to auditory sequences let me test your memory for sequences you think you could remember this sequence okay this is eight I'm sorry 16 tones long 16 tones that can be a or b high or low let me have you listen to it again okay I think you will agree that it's almost impossible to remember at least you will need many trials before you can remember it now let's listen to this one you think you could repeat it the the evidence is that you can maybe you'll need two trials but not more because this is highly regular and maybe here's another one okay Your Capacity to remember these sequences is again proportional to their regularity we've tested this in two different ways one is just to ask subject for subjective complexity rating so just rate the complexity as you feel but the other is to ask for an objective measure of complexity can you detect whether there is a violation one of these tones is wrong can you detected and this is the percentage of detection and combined also with the time that it takes you to detect so remarkably the very same measure of complexity that we used for the geometric sequences can be used for the auditory sequences it's a cross modal notion of regularity the only thing that you need to capture uh what's Happening Here is a notion of nested repetition there is repetition of repetition of repetition and you can compress the sequence by understanding how this repetitions behave do you switch from A to B or not once you have that the length of the compress compressed expression is a direct predictor of how complex the sequence is to remember so there is something in the mind of humans that compresses the information can monkeys do this at all the evidence is that they count in fact they may be missing some of the most basic mechanism in order to integrate information as we do in humans here's a nice experiment by ding War when he was in the lab um well the idea was we should ask how do subjects really understand the local Global sequence that I showed you earlier so let me have you listen to the pattern okay I hope you perceive that even though there is variation now between the tones there's a lot of regularity it's all always three identical sounds and then there's a different one can monkeys can monkeys understand that do they have something in mind just like us they would say it's three tones X and then another tone y um this seems to be a sort of minimal expression that humans will use in order to compress this information do monkeys have this at all in order to test it we had several violations you can change to a new uh set of tones but still keep the same rule so it's new example of the same role or you can violate the number you can have only two or maybe six items or you can violate the sequence itself you expect that the last one is going to be different but it's not or you can have double deviant double the violation so it's the wrong number and also the last item is not is not right we scan monkeys and we scan humans during this task it's a complex set of results while I will summarize it very briefly this is the activation to the violations in number and this is the activation evoked to the violation of sequences the last item R and this is the intersection in yellow you can see that the monkey has both he has regions such as the int paral cortex which we know cares about number and reacts to number violations and we find that in the human as well we have a number sense ability to react to number we also have responses to uh violations of sequences in the monkey you can see that in Green in particular in inferior frontal cortex what we don't have in the monkey is the areas in in yellow in humans there is a whole set of areas in inferior frontal gy Brokers area here and in the psts in both hemispheres that react to the violation of both rules and that seem to encode a single formula for the entire rule that the monkey or the human has been remembering and we really see this as a strong dissociation humans show similar pattern of activity for number sequence change as if they the single representation sorry about that monkey show negative correlation so if a voxel activates for number it does not activate for sequence so monkeys don't have the integrated representation that we suggest is present in humans and may be present in fact around broker's area around the inferior prefrontal cortex in several areas of human prefrontal cortex we find evidence for this integration according to rules so monkeys May simply not be able to integrate the information consciously in the same way that we do if I still have a few minutes I would like to show you this in yet another form which I think is a nice challenge for artificial intelligence um this is a drawing undoubtedly from humans not from any ape not any not from any nonhuman ape uh you can see the evidence of drawing an animal but you can also see geometrical shapes here a rectangle or a series of dots is this unique to humans indeed would a monkey ever understand this sort of shape uh I like this joke which appeared in the New Yorker here you have these humans watching a hunting scene and they say we have to record this or no one's going to believe us because what they see is not the actual animal they see the shapes they see the abstraction just like the artist that Dre this in Las can we test this can we test the idea that even when we look at a very simple geometrical form we are compressing it using its regularity we are seeing regularities so this is the work of Matas saer in the lab um what we do here is we present you a very simple display like this one here and you have to spot the OD ball the outlier and I hope you've seen that this one is the OD ball because these are five rectangles and this one is not a rectangle we do this for every possible shape that is displayed here from highly regular square or rectangle to completely irregular shapes that have no way to be compressed rather than remembering the specifics of the particular shape and we find that regularity has a dramatic effect on subject's ability to detect the outlier this is another example here it's a bit harder to detect the outlier I hope for you until you find this one as a displaced Corner we always displace the corner this is important by the same amount so it's not the amount of displacement that counts it's the regularity of the shape the more regular the shape is the better you are so you're super good at detecting violations of a square or rectangle you are very very poor at detecting violations of an irregular shape and the other ones they fall in between as a function of the degree of regularity so the humans may have this special notion of a square as something which is regular particular it is a complete human Universal I don't think I have time to explain to you all the replications of these effects in different paradigms just subjective ratings again we have a notion of introspection we have a conscious introspection for the complexity of these shapes we can even present them as sequences it doesn't have to be a shape so that's a little bit of a challenge for let's say CNN's uh convolutional neuron networks because here the humans resist to the presentation of the shapes as a sequence of the Four

Corners um but even preschoolers or the HBA here adult people who have very little access to education they live in Namibia outside of the normal Western environment they still show this regularity effect so it's a complete human Universal in our hands at least doesn't depend on education or at least uh it's present already in a sort sort of simple form in the absence of education so what about monkeys we we were able to test it in baboons this is a wonderful facility in the south of France which Joel fago running this facility where the baboons can access this booth at any given time and they can get video games this could be a 2 in the morning you have a monkey entering and he's recognized by the system and he has access to a video game and the video game is of course the one that we designed uh we don't start with the I'm getting angry okay Mr Panda I'm closing so I think you gu where I'm got getting baboons can learn the outlier Tas they can detect the ODB flower here they can generalize to all of these stimuli so they are able to learn the outb tars but what they don't do is show a shape regularity they are completely from and you see that are essentially flat they detect the outliers regardless of whether it's a square or a or very irregular shape which serves as a base and we claim that CNN capture very well in multiple regression the behaviors of baboons but that they don't capture what's happening in humans not at all and that a completely different model is needed in order to capture what's happening in humans a model which lists the internal properties of the shapes I think I will just go to my conclusion I think that human consciousness is special of course there is a conscious Global workspace in all primates and perhaps in many other animal species there's a beautiful paper by Andreas Neer on corvids showing the same sort of threshold and ignition in uh the crown I think we also share with many other primates the same Core Concepts for instance the concept of number of object but also of space of person and so on the basic concepts are there in other primates but maybe according to this new hypothesis only humans possess an internal language of thought that discretizes first of all the concepts into mental tokens or symbols and combines them recursively into mental programs whose complexity determine whether we're going to be able to remember them or not and my argument is that something happened during omanization our brain architecture must have changed in a way that facilitates the acquisition of recursive languages and this happens in several brain networks language areas but also Mass responsive areas or theory of Mind regions giving rise to the extraordinary complexity of human conscious experiences this second part here what's special to the human brain I would argue is not yet well captured in artificial intelligence it's more like system two that Joshua benio was discussing so we might have a discussion about that with that I will leave you with these two books and thank you very much for your attention hello hello hello yes yes thank you so much was an impressive talk um as expected that was wonderful mind-blowing a lot of beautiful data the cognitive experiments are just so mind-blowing and amazing um and it nicely compliments um all of the work we've been looking at that today um we do have time for questions and we have to take questions there just way so many questions so I'm just going to pull up he the questions and see those that have been uploaded most and maybe we can go through a few of them and again we'll have time later in the panel so I'm gonna ask first so the first question um You probably see it also stand on the ask question tab the first one is by uh a Miller who's asking uh in the global Network space in the gmw Theory does Consciousness require um control oriented interceptive uh inference um as advocated by Anil in his talk this morning so the way to incorporate interceptive inference yeah I I don't think so I I I think that there is a lot but I I may not fully understand the concept that Anil has been proposing but I I I think the theory that I'm proposing is that we can be conscious of many different things and uh this does not necessarily require um any sense of self or any any notion of interception so on all it requires is an internal system for broadcasting information can be very abstract so think of what you are doing when you're doing mathematics um but I'm not sure I fully understand also the particular proposal here so fair enough maybe we should discuss that later yeah yeah we can get back to it so um I'm G to take a question by um Andy disra who's asking how do you address the work of Michael pittz Steve Hillard and others that claims that the p3b is postperceptual rather than perception actually per yeah I think it was a little bit wrong to claim that one particular wave which of course varies from Paradigm to Paradigm also is can be a universal signature of Consciousness so so that has been clearly refuted I think what is still clear is that if you have a novel stimulus one which is not expected then the PB is not a b sign of consciousness um but as soon as the stimulus becomes habitual expected uh you don't necessarily get it um and also if the stimulus does not require a lot of postprocessing as is being said in the question then it can be severely reduced that does not I think detract from the fact that you have ignition the ignition phenomenon is still present as I showed I think especially in the new neural recordings in the prefrontal cortex you have very brief activation but still crossing the prefrontal cortex and if you don't need to hang on to it because you're just passively looking then you get it and then it's gone it flies by basically so it will not create this very Global wave that you can detect from the outside which called the p3b the p3b will be seen if you are surprised or if you have to actively report the St um thank you I'm going to switch to a question by Andrea brelli so Andrea is asking um so from Mar he's asking differences in movements be it sads or micro sads may exist between conscious so seen and unconscious unseen processing of visual stimuli uh which may lead to different responses for example with Meg in the prefrontal cortex starting like from 400 milliseconds so the question is would you exclude such potential confounding effects we typically do uh we typically always record ey movements in Meg it's also the case of course in the monkey experiments and we can analyze show that there is no detectable difference now you could always argue that there are you know micro Cades that are still present of course below the resolution of the particular scanner but I think there are enough paradigms now that we can exclude this interpretation in particular in the auditory paradigms I see no reason why there would be a contamination with the movements okay so maybe one last question here before we move on later on to our panel discussion this is a question by Hikaru uh to jimura and basically it relates to u to linking the uh Global neural workspace to default mode Network so question is what is the core difference between uh the global neural workspace and the default mode Network especially the advantage of the of the global neural workspace to explain mechanisms of Consciousness or awareness that's an interesting question uh first of all I think that in science it's often the case that we name same things with different names and we see the bridges later on so this could be a case um however I think that the global neuronal workspace is broader than the particular default mode Network which typically pointed out in some studies for instance um you get a different network within the global workspace you get different topologies of the network depending on the contents of what's being accessed by the subject and default mode Network seem to relate more to verbal contents perhaps although it's bilateral so in the right emisphere is more than that but we can also get slightly different network if you think about mathematical Concepts instance and then it's a clearly different network so I think that uh this relates to what I said in the introduction there a style of processing which is associated with conscious processing but it's not necessarily a fixed set of areas that are always coming up when you're conscious it's the style of activation ignition the fact that the subset of prefrontal neurons are being activated and the others are being silenced which tells us that the subject was conscious of a piece of information um awesome so I will take one last question because this one's the most outvoted one um so you can probably see it um at the very top um of the list uh so the question is in the global neural workspace um an important role is ascribed to the prefrontal cortex but the question is what about the hippocampus which sits on top of the cortical sensory hierarchy and has important roles in obviously Association memory of course and space and time so how how do you see hyppocampus this is a great question and very important one there is the hypoc campus there is also the thalamic nuclei that are connected with the cortex and um of course in human we tend to study mostly the cortical responses and also in the monkey now that we have access to PFC recordings um but I completely agree with the notion of the question that this is a this is a linked system and I think it's often the case that what gets recorded in episodic memory has to be conscious first so uh I think there's a sense in which it's highly likely that what gets stored in the hypoc compus in episodic memory is was first a cortical content that was ignited and in this sense that would be a very important relation between the two it works also in the opposite direction when you access consciously a memory you can search your memory and suddenly it pops into your Consciousness this is ignition it comes from the hypoc compus it doesn't come from the external world but it comes from the hypo campus being able to replay and eventually ignite an entire assembly in prefontal cortex and related areas so once again this is not a localist theory I think because it is a global workspace it needs a distributed system and I don't like uh the idea of reducing it to just prefrontal cortex I think it's very likely that the hypoc compus plays an important role now we know that if you have a legion in the hypoc campus you can still be conscious again this points to the idea that this is a distributed system it's not one node that will knock it off but still dynamically these nodes tend to come up together when they activated great thank you so much Stan for all these enlightening answers to these questions and to the wonderful talk once again um we could go on forever with these questions there are many more popping up but uh in the interest of time and because we have the panel uh to be able to continue this discussion I think we will move on so thanks once again so please everybody join me in thanking s Doan for this amazing keynote lecture at Main 2020 and we will be closing this session in a few maybe in about a minute or so before we switch on to the next one which is going to be the panel discussion um and just one note because we are many people meeting we cannot have the panel discussion at least from a speaker perspective in crowdcast you all stay on crowdcast but we will be moving to zoom and will be rebroadcast through crowdcast so you don't have to worry about anything just the speakers this is a reminder to the speakers joining the panel you now need to close your your uh crowdcast window and and switch to the zoom link that we sent you but for everybody else just stick around everything is going to continue happening on uh crowdcast uh fingers crossed everything goes well I hope so thank you once again and one last comment that I'd like to make this is out there for the students newcomers of the field and people who have heard a lot about Consciousness this morning um and who might not have the background and find it interesting uh or who have some background but would like to know more um I would just like to point out there's going to be um an introductory uh lecture tomorrow morning because tomorrow is our introductory courses lectures uh of Main 20120 and there there's one tomorrow uh which is going to be uh given by Stephanie so Professor Stephanie blam morz from mild who's been working on Consciousness for quite a while um and has a strong expertise in the field so she will be giving a talk on the introduction to um the basics of studying uh Consciousness uh with Neuroscience so the neuroscientific approaches to the study of Consciousness an introduction that's going to be uh actually the opening lecture tomorrow morning at um 8 or 8:20 I think it's 8:20 tomorrow morning so make sure you don't miss that um uh and so that's all I wanted to tell you right now um we'll be closing the session and um meeting up again in a couple of minutes for the panel discussion and just to let you know the panel disc discussion is going to be when we start it uh you will see there will be some questions that will be sent in the poll so down at the bottom you you'll see appearing a button where you can press and we've prepared a couple of questions because we want to ask you make this more interactive so there'll be two or three slightly provocative or confusing questions about Consciousness uh so as soon as you see those question the poll appear please answer those questions and we'll have a look at at your answers thanks once again and thank you to all the speakers e